Chapter 9

Environmental Microbiology

ERMVCX

Chapter Outline

9.1 Interrelationship with other Fields

9.2 Air Microbiology

9.3 Microbiology of Water

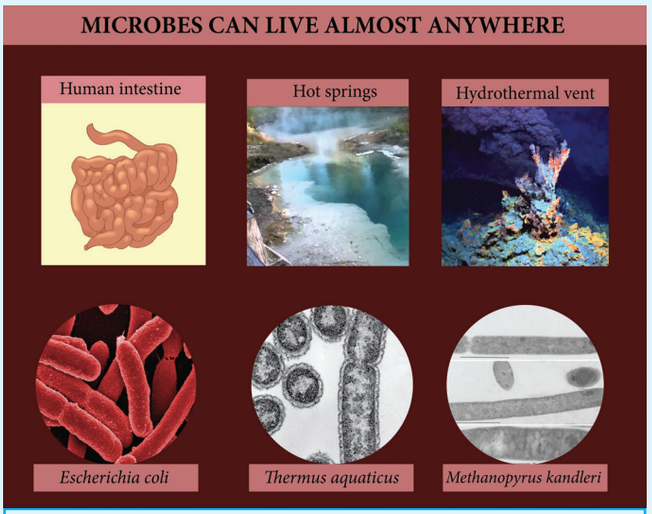
9.4 Water Pollution and Microbial Contamination

9.5 Sewage Treatment

9.6 Recycling of Treated Sewage

9.7 Composting

9.8 Biogas Production



Microorganisms are found in every corner of the earth from miles below the surface to boiling hot springs to Antartic ice. Every ecosystem on earth contains microorganisms thay occupy unique niches based on their specific metabolic properties.

Learning Objectives

After studying this chapter the student will be able,

- *To gain knowledge of various layers of atmosphere and micro fauna of air.*
- *To understand air pollution and air borne diseases.*
- *To learn water borne diseases and water treatment procedures.*
- *To know Eutrophication.*
- *To know composting techniques.*
- *To gain knowledge of biogas production.*

Environmental Microbiology is the field of science that examines the relationship between microorganisms and their biotic and abiotic environments. Microorganisms in the environment are diverse in origin and ubiquitous. Environmental Microbiology involves the study of the applied effects of microorganisms on the environment and on human activity, health and welfare.

It was in the 1970s that a new area of Microbiology emerged and developed into the field of Environmental Microbiology. The initial focus was on water quality pathogens in the environment in the context of public health safety. The developing field of environmental microbiology expanded to several other areas of applied research. These include microbial interactions with chemical pollutants in the environment and the use of microorganisms for resource production and resource recovery.